
Group A

Assignment No: 9

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1. Title of the Assignment: Data Visualization III Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as:
2. List down the features and their types (e.g., numeric, nominal) available in the dataset.
3. Create a histogram for each feature in the dataset to illustrate the feature distributions.
4. Create a box plot for each feature in the dataset. 4. Compare distributions and identify outliers.

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1. Theory:

As the "age of Big Data" kicks into high gear, visualization is an increasingly key tool to make sense of the trillions of rows of data generated every day. Data visualization helps to tell stories by curating data into a form easier to understand, highlighting the trends and outliers. A good visualization tells a story, removing the noise from data and highlighting useful information.

However, it's not simply as easy as just dressing up a graph to make it look better or slapping on the "info" part of an infographic. Effective data visualization is a delicate balancing act between form and function. The plainest graph could be too boring to catch any notice or it make tell a powerful point; the most stunning visualization could utterly fail at conveying the right message or it could speak volumes. The data and the visuals need to work together, and there's an art to combining great analysis with great storytelling.

2. Different types of visualizations

When you think of data visualization, your first thought probably immediately goes to simple bar graphs or pie charts. While these may be an integral part of visualizing data and a common baseline for many data graphics, the right visualization must be paired with the right set of information. Simple graphs are only the tip of the iceberg. There's a whole selection of visualization methods to present data in effective and interesting ways.

3. **General Types of Visualizations:**

1) Chart: Information presented in a tabular, graphical form with data displayed along two axes. Can be in the form of a graph, diagram, or map. Learn more.
2) Table: A set of figures displayed in rows and columns. Learn more.
3) Graph: A diagram of points, lines, segments, curves, or areas that represents certain variables in comparison to each other, usually along two axes at a right angle.
4) Geospatial: A visualization that shows data in map form using different shapes and colors to show the relationship between pieces of data and specific locations. Learn more.
5) Infographic: A combination of visuals and words that represent data. Usually uses charts or diagrams.
6) Dashboards: A collection of visualizations and data displayed in one place to help with analyzing and presenting data

4. **More specific examples**

1) rea Map: A form of geospatial visualization, area maps are used to show specific values set over a map of a country, state, county, or any other geographic location. Two common types of area maps are choropleths and isopleths.
2) Bar Chart: Bar charts represent numerical values compared to each other. The length of the bar represents the value of each variable.
3) Box-and-whisker Plots: These show a selection of ranges (the box) across a set measure (the bar).
4) Bullet Graph: A bar marked against a background to show progress or performance against a goal, denoted by a line on the graph.
5) Gantt Chart: Typically used in project management, Gantt charts are a bar chart depiction of timelines and tasks.
6) Heat Map: A type of geospatial visualization in map form which displays specific data values as different colors (this doesn’t need to be temperatures, but that is a common use).
7) Highlight Table: A form of table that uses color to categorize similar data, allowing the viewer to read it more easily and intuitively.
8) Histogram: A type of bar chart that split a continuous measure into different bins to help analyze the distribution.

9) Pie Chart: A circular chart with triangular segments that shows data as a percentage of a whole.
10) Treemap: A type of chart that shows different, related values in the form of rectangles nested together.

5. Algorithm:-

[1]:
```
import pandas as pd
import seaborn as sns
```

In [2]:
```
data=pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\10\Iris.csv")
```

In [3]:
```
data.head()
```

Out[3]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [4]:
```
data.describe()
```

Out[4]:

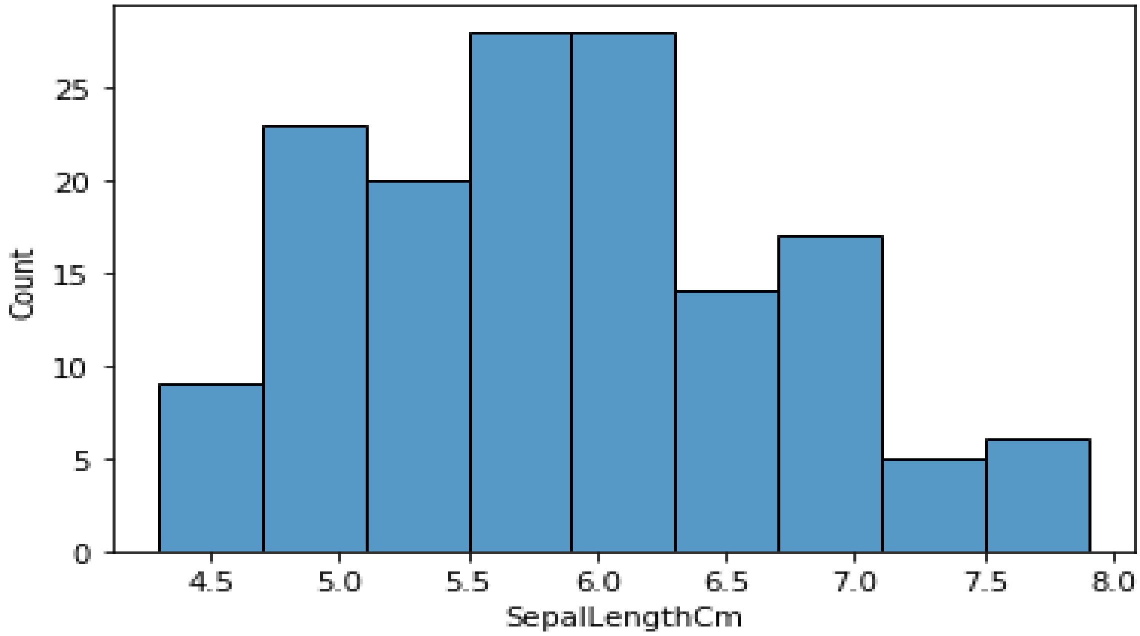
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

In [5]:
```
data.info()
```

```
<class
'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
Id                    150 non-null int64
SepalLengthCm 150 non-null float64
SepalWidthCm  150  non-null  float64
PetalLengthCm  150  non-null  float64
PetalWidthCm  150  non-null  float64
Species    150 non-null object
dtypes:  float64(4),  int64(1),  object(1)
memory usage: 7.2+ KB
```

In [6]: sns.histplot(data=data, x="SepalLengthCm")

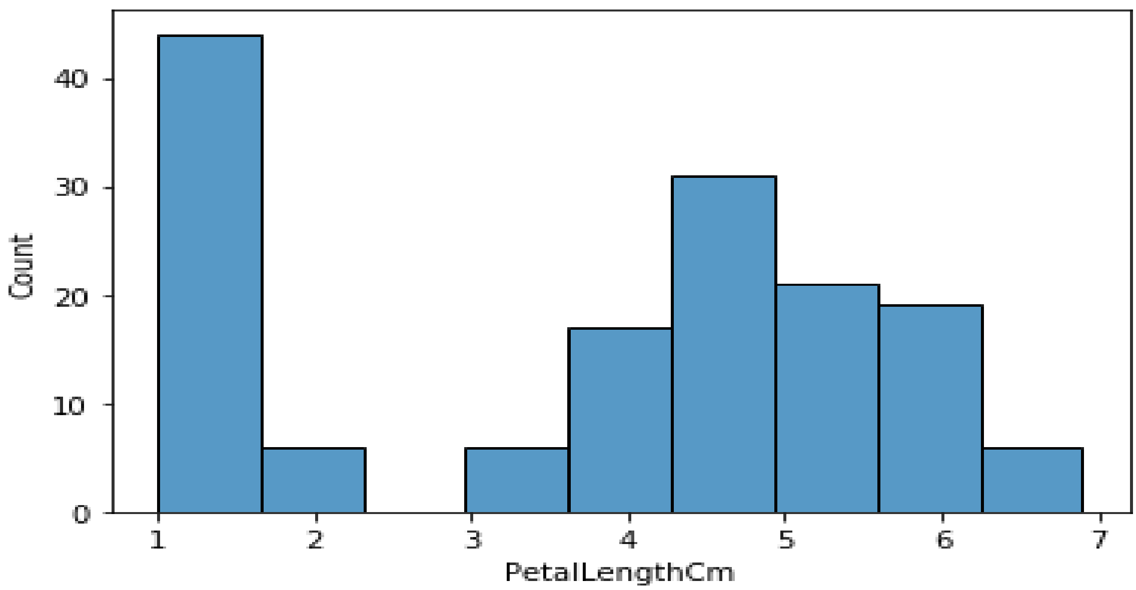
Out[6]: <matplotlib.axes._subplots.AxesSubplot at 0x2274e737808>



In [7]:

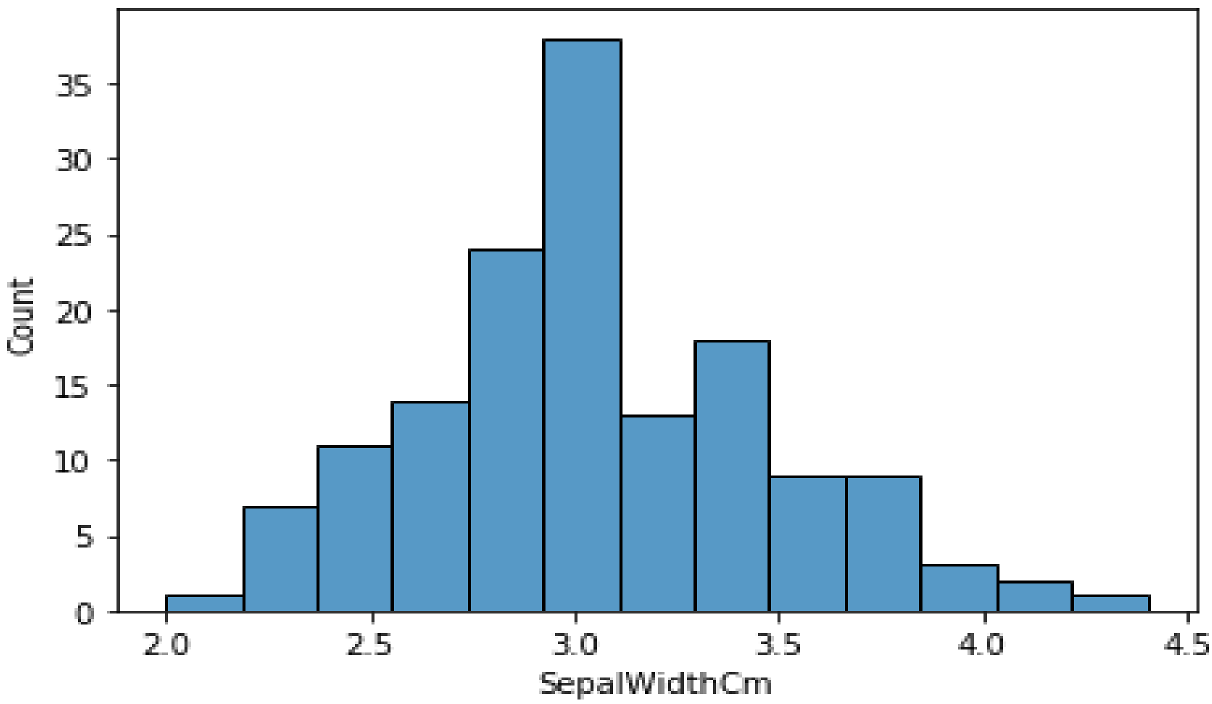
```
sns.histplot(data=data, x="PetalLengthCm")
```

Out[7]: <matplotlib.axes._subplots.AxesSubplot at 0x2274eab8888>



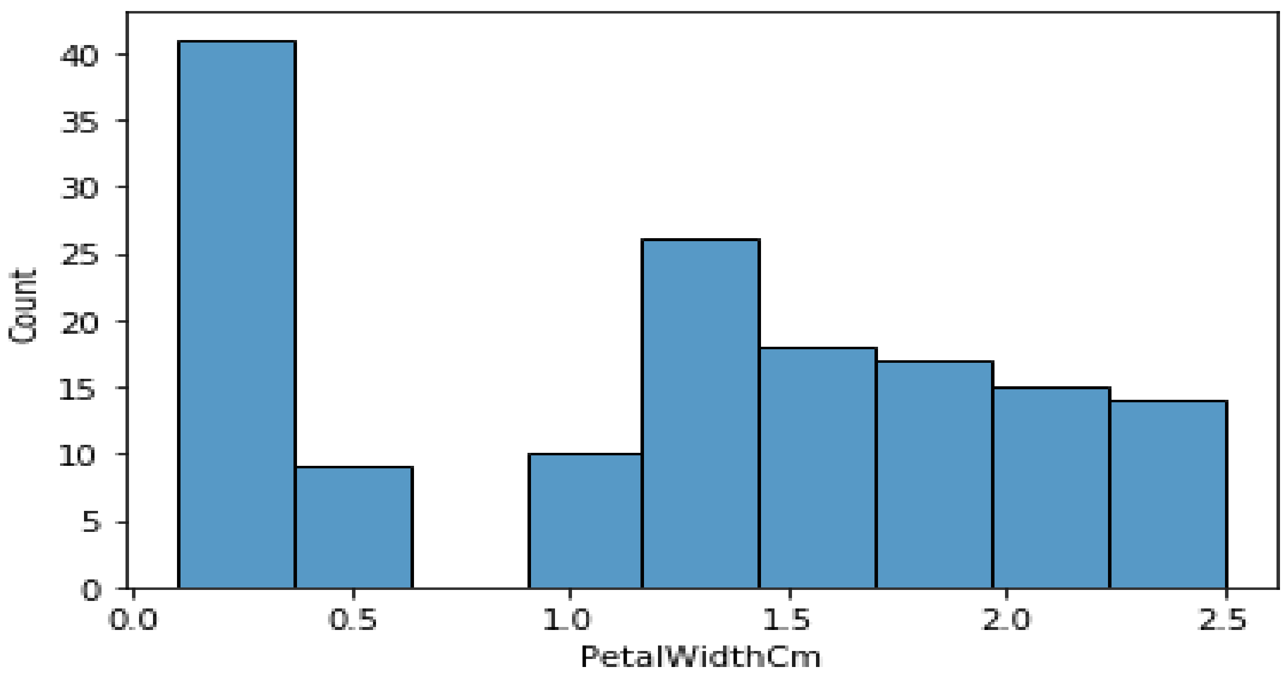
In [8]: `sns.histplot(data=data, x="SepalWidthCm")`

Out[8]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fb0c148>



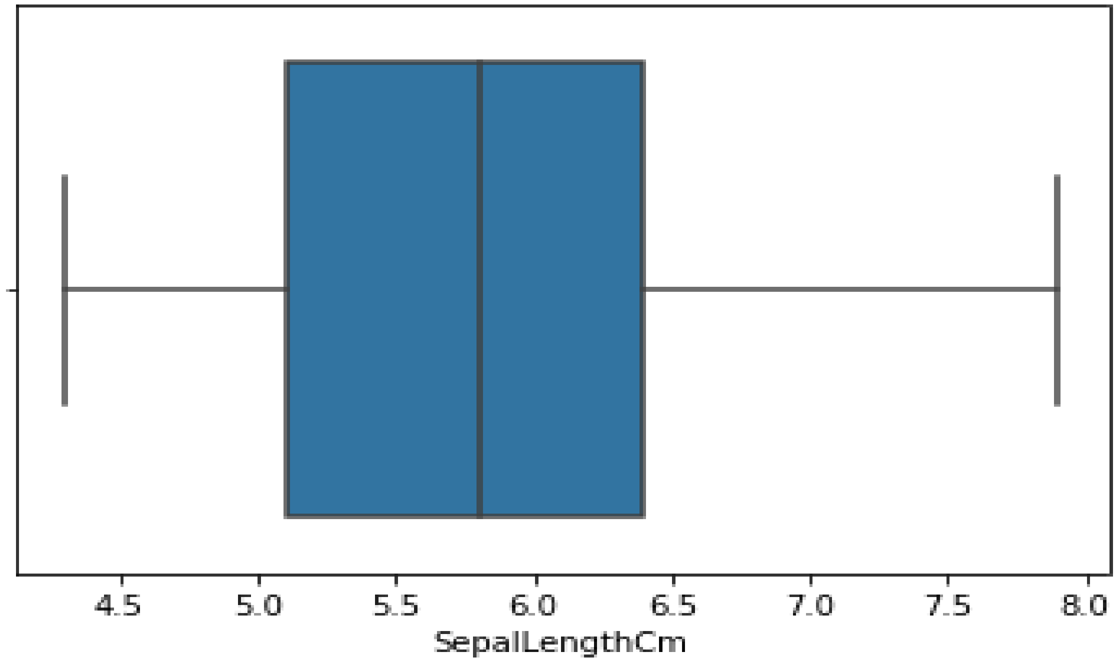
In [9]: `sns.histplot(data=data, x="PetalWidthCm")`

Out[9]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fbb9088>



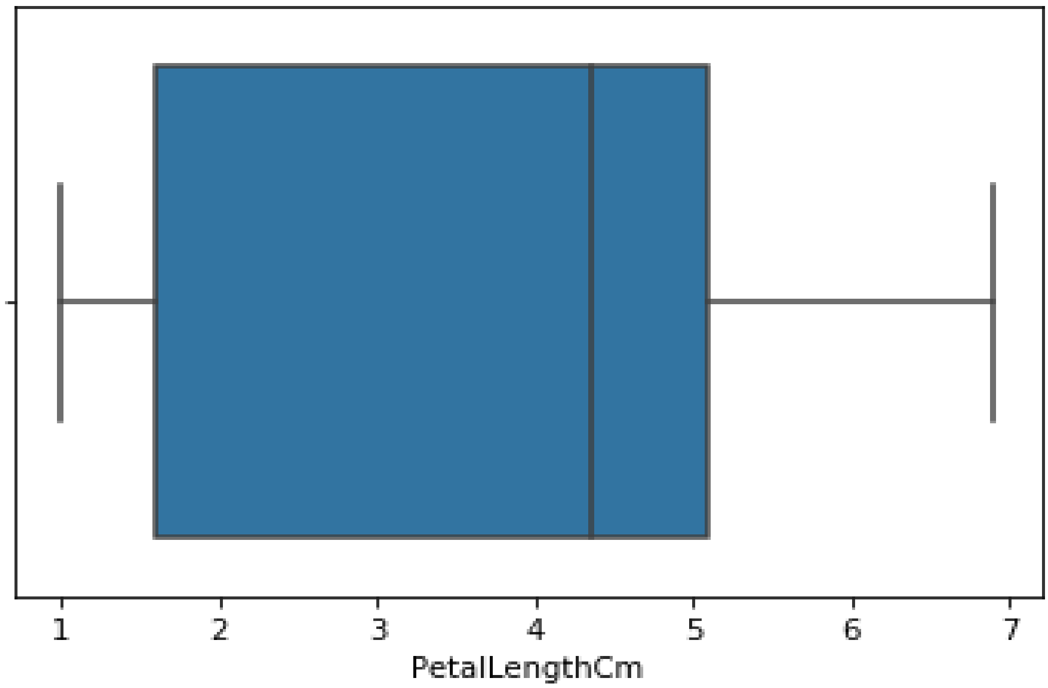
```
In [10]: sns.boxplot(x="SepalLengthCm", data=data)
```

Out[10]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fc47d88>



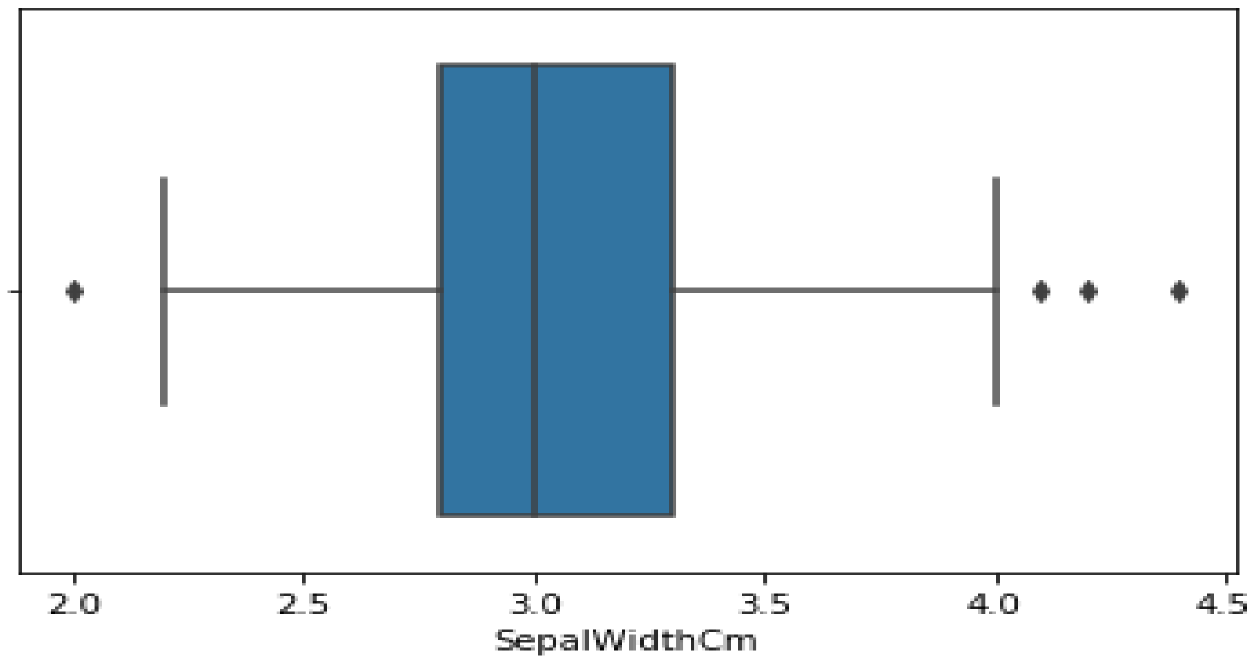

```
In [11]: sns.boxplot(x="PetalLengthCm", data=data)
```

Out[11]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fc977c8>



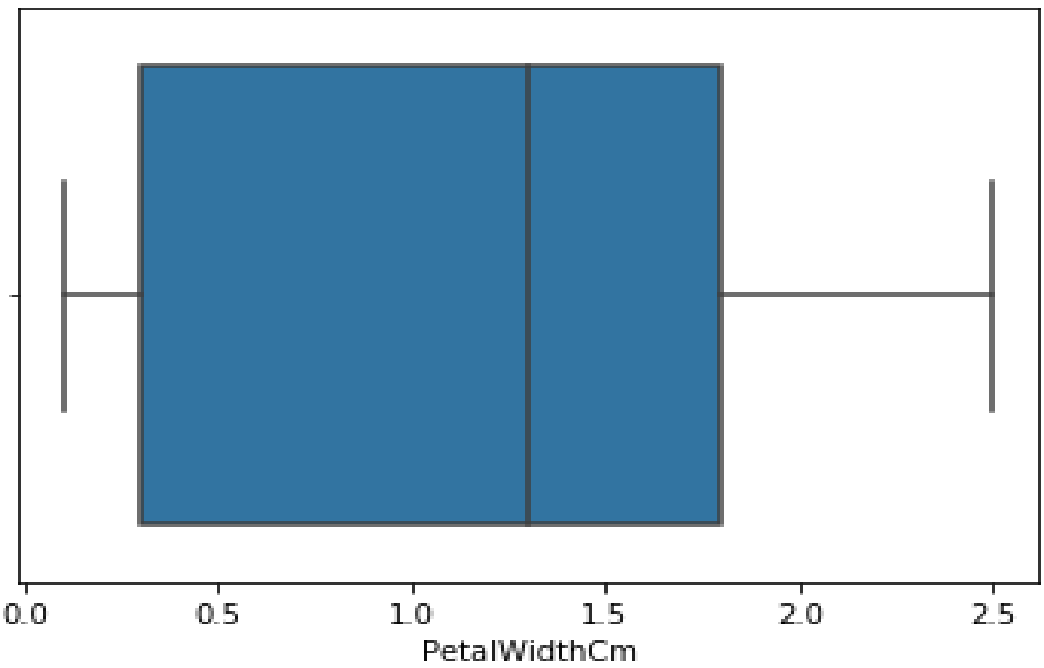
```
In [12]: sns.boxplot(x="SepalWidthCm", data=data)
```

Out[12]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fd2a3c8>

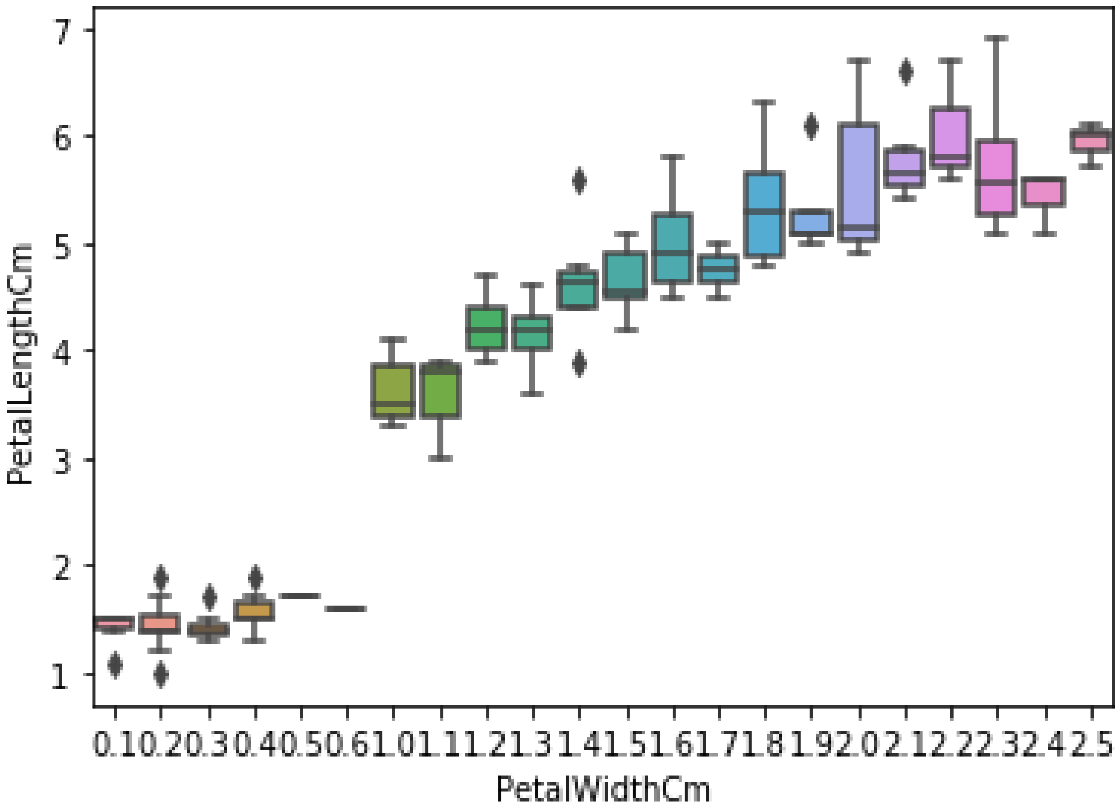



```
In [13]: sns.boxplot(x="PetalWidthCm", data=data)
```

Out[13]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fc58548>



Out[14]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fdd4648>



In [15]:
```
sns.boxplot(x="SepalWidthCm", y="SepalLengthCm", data=data)
```

Out[15]: <matplotlib.axes._subplots.AxesSubplot at 0x2275004c188>